

For the system $dz/dt = f(z)$, a coordinate transformation $z \mapsto \tilde{z} = Rz$ results in

$$\frac{d\tilde{z}}{dt} = R \frac{dz}{dt} = Rf(R^{-1}\tilde{z}) = Rf(R\tilde{z}),$$

since $R^{-1} = R$.

A change of time $t \mapsto \tau = -t$ results in

$$\frac{dz}{d\tau} = \frac{dt}{d\tau} \frac{dz}{dt} = -f(z).$$

In other words, for a reversible vector field, the coordinate transformation $z \mapsto Rz$ is equivalent to the change of time $t \mapsto -t$.

3.6.2 Time-Reversal Symmetry and Symmetric Numerical Methods

The flow map $\mathcal{F}_t(z_0)$ gives a solution of the differential equation, thus for a time-reversible system, we have, with $z(t) = \mathcal{F}_t(z_0)$,

$$\frac{d}{dt} \mathcal{F}_t(z_0) = \frac{dz}{dt} = -Rf(R\mathcal{F}_t(z_0)).$$

Hence

$$\frac{d}{dt} R\mathcal{F}_t(z_0) = -f(R\mathcal{F}_t(z_0)),$$

and if we define $\tilde{z}(t) = R\mathcal{F}_t(z_0)$ then $\tilde{z}(t)$ satisfies the differential equation

$$\frac{d\tilde{z}}{dt} = -f(\tilde{z}).$$

From this we see that if we start from a point z_0 and integrate forward in time (i.e. apply $\mathcal{F}_t$) then apply the involution R , we get the exact same result as if we start from $\tilde{z}(0) = Rz_0$ and integrate *backward* in time (i.e. apply $\mathcal{F}_{-t}$). In sum, since z_0 is an arbitrary point,

$$\mathcal{F}_{-t}(Rz) = R\mathcal{F}_t(z).$$

For the flow map, $\mathcal{F}_{-t} = \mathcal{F}_t^{-1}$ so we can recast the identity as

$$R \circ \mathcal{F}_t \circ R \circ \mathcal{F}_t = \text{Id}, \quad (3.14)$$